

Mitotic Image Classification in Glioma Using Attention-Enhanced CNNs

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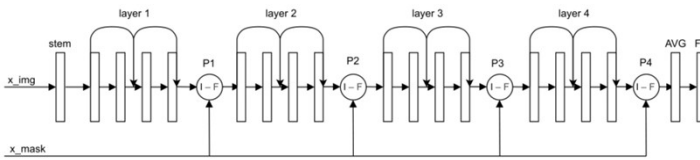
Introduction

Detecting mitotic cells is **vital**, but manual counting is **slow** and **subjective**. Current AI automates this but relies on **expensive**, heavy **transformer-based** models. Our goal is to design an **accurate**, **lightweight** AI that is accessible for **everyday clinical use**.

The Objective

Evaluate **8 modern attention mechanisms** across **ResNet18** and **MobileNetV3** backbones and propose a highly efficient "**Guided Attention**" and "**Segmentation-guided Spatial Attention**" model using **binary** Region of Interest (**ROI**) masks to rival heavier attention blocks.

Proposed architectures



A scheme of ResNet18 with our added I-F attention blocks

Interpolate-Fuse (I-F) Blocks: Binary masks are **resized** and **normalized** into a **soft-attention distribution** A_{ij} . This computes a global context vector v **fused** back into key residual stages (**P1-P4**) - our "**Guided Attention**"

$$A_{ij} = \frac{\exp(S_{ij})}{\sum_{u,v} \exp(S_{uv})} \quad v = \sum_{i,j} A_{ij} F_{i,j}$$

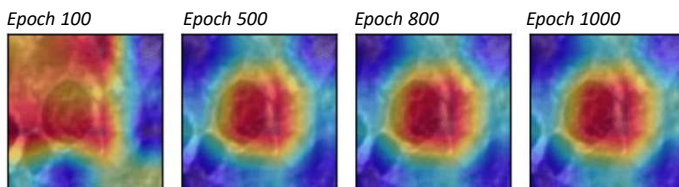
Interpolate-Convolve (I-C): Our "**Segmentation-guided Spatial Attention**" uses a small **convolution** and a **sigmoid** to generate a 0-to-1 spatial mask.

$$M = \sigma(\text{Conv}_{3 \times 3}(\text{Interp}(S))) \quad F' = F + F \odot M$$

GradCAM Explainability

Medical AI requires strict **explainability**.

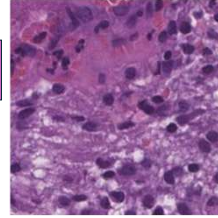
GradCAM saliency heatmaps verify our models focus on the **correct** cellular features, converging to desired **ROIs** on **par** with **transformer-inspired OutlookAttention**.



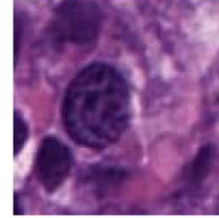
GradCAM visualizations for **res_guided_2** model at various training epochs

Dataset

Trained on **2908 augmented, 100×100** pixel samples extracted from the **Glioma-MDC 2025** challenge dataset. Samples focus specifically on individual ROIs in **H&E-stained human glioma tissue**.



Dataset image



Extracted Cell



The Cell's ROI Binary mask

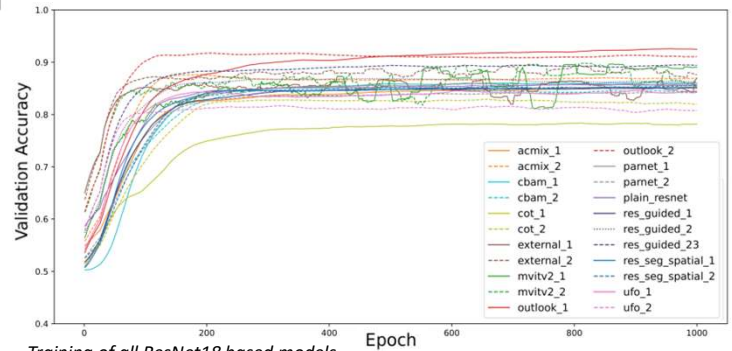
Training

Validation: **5-fold cross-validation**.

Optimization: **1000 epochs** (batch 128), **Adam**, **CE-Loss**

Learning Rates: 5×10^{-7} **ResNet18**, 5×10^{-6} **MobileNetV3**

Convergence: **ResNet18** peaked between epochs **100-500**



Training of all ResNet18 based models

Results

Model	Mean			
	Accuracy	Precision	Recall	F1 ± SE
outlook_2	0.93	0.92	0.93	0.93 ± 0.005
outlook_1	0.92	0.95	0.90	0.92 ± 0.003
mvitv2_1	0.91	0.91	0.91	0.91 ± 0.007
res_guided_2 (Ours)	0.89	0.87	0.93	0.90 ± 0.003
mvitv2_2	0.89	0.89	0.90	0.90 ± 0.003

OutlookAttention achieved the **highest overall mean F1** score of **0.93**

Our **res_guided_2 ResNet18** with **Guided Attention** at P2 reached an **F1** of **0.90** and a **0.93 recall** rate.

Conclusion: Our model **matches OutlookAttention** on the most **critical** medical metric - **recall** - proving that simple, hand-drawn masks can effectively rival much heavier attention blocks.

GitHub: <https://github.com/Dosmar/MitoticFiguresDetectionWithAttention>